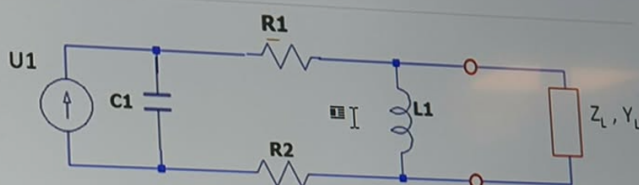


UKEL - KOL 1 - 2022 gr.2

Zadanie 1.

Zad.1. Należy wyznaczyć optymalną impedancję i admitancję obciążenia Z_L i Y_L oraz moc dostarczoną do obciążenia w tych warunkach dla układu.

$U_1(t)$	$50 \sin(\omega t)$ V
f	$10/2\pi$ MHz
C_1	500 pF
L_1	5 μ H
R_1	10 Ω
R_2	40 Ω



$$U_1 = -50j \text{ V}$$

$$\omega = 2\pi f = 2\pi \cdot \frac{10}{2\pi} \cdot 10^6 = 10^7 \frac{\text{rad}}{\text{s}}$$

$$Z_1 = R_1 + R_2 = 50 \Omega$$

$$Z_{L1} = j\omega L_1 = j \cdot 10^7 \cdot 5 \cdot 10^{-6} = 50j \Omega$$

$$Z_2 = \frac{Z_1 \cdot Z_{L1}}{Z_1 + Z_{L1}} = \frac{50 \cdot 50j}{50 + 50j} = 25 + 25j \Omega$$

$$Z_L = Z_2^* = 25 - 25j \Omega$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{25 - 25j} = \frac{1}{50} + \frac{1}{50}j \text{ S}$$

$$E_{th} = U_{L1} = U_1 \cdot \frac{Z_{L1}}{R_1 + Z_{L1} + R_2} = -50j \cdot \frac{50j}{50 + 50j} = 25 - 25j \text{ V}$$

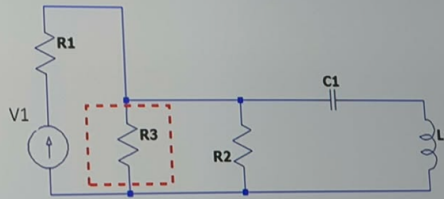
$$I_x = \frac{U_x}{Z_L} = \frac{25 - 25j}{25 - 25j} = 1 \text{ A}$$

$$P = \frac{|E_{th}|^2}{8 \cdot \text{Re}(Z_L)} = \frac{|25 - 25j|^2}{8 \cdot 25} = 6,25 \text{ W}$$

Zadanie 2.

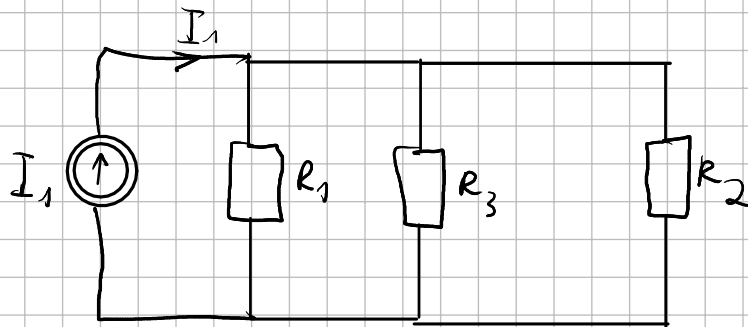
Zad.2. Należy wyznaczyć: $i_{R2}(t)$, $u_{R2}(t)$, oraz moc wydzielaną w rezystorze $R2$.

V1	$V_1(t) = 20 + 30\sin(\omega t) + 20\cos(2\omega t)$ [V]
f	$100/(2\pi)$ MHz
C1	1 nF
L1	100 nH
R1	10 Ω
R2	20 Ω
R3	20 Ω



1) DC

Zamiana na źródło prądowe



$$V_1 = 20 \text{ V}$$

$$I_1 = \frac{20 \text{ V}}{10 \Omega} = 2 \text{ A}$$

$$R_{13} = \frac{10 \cdot 20}{30} = \frac{200}{30} = \frac{20}{3} \Omega$$

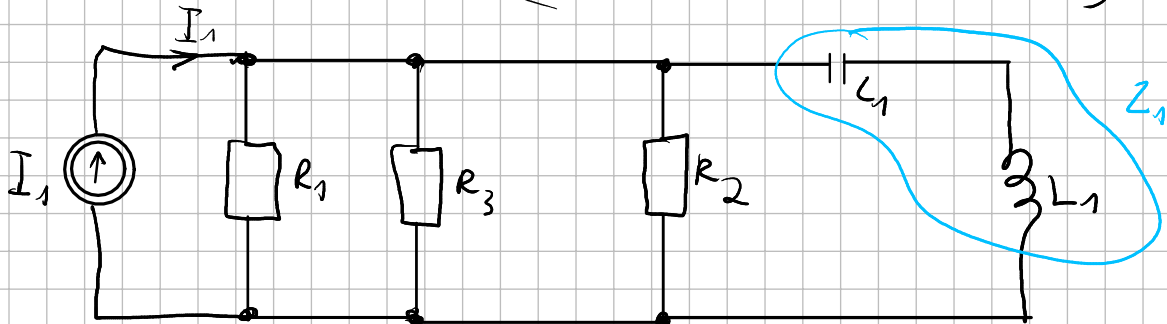
$$i_{R2} = I_1 \cdot \frac{R_{13}}{R_{13} + R_2} = 2 \cdot \frac{\frac{20}{3}}{\frac{20}{3} + 20} = 0,5 \text{ A}$$

$$u_{R2} = i_{R2} \cdot R_2 = 0,5 \text{ A} \cdot 20 \Omega = 10 \text{ V}$$

2) $V_{12}(t) = 30 \sin(\omega t)$

$$V_{12} = 30 \angle 0^\circ \text{ V}$$

$$\omega = 2\pi f = 2\pi \cdot \frac{100}{2\pi} \cdot 10^6 = 100 \cdot 10^6 \frac{\text{rad}}{\text{s}}$$



$$Z_1 = Z_{C1} + Z_{L1} = \frac{1}{j\omega C_1} + j\omega L_1 = \frac{1}{j \cdot 100 \cdot 10^6 \cdot 10^{-9}} + j \cdot 100 \cdot 10^6 \cdot 100 \cdot 10^{-9}$$

$$Z_1 = 0$$

C_1 oraz L_1 szeregowo zachowują się jak
zwarciła ($Z_1=0$).

$$i_{r22} = 0$$

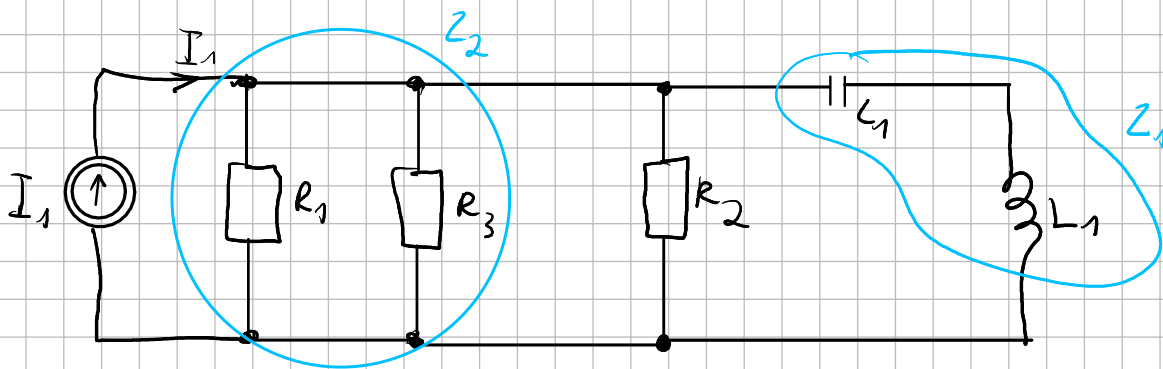
$$u_{r22} = 0$$

3)

$$V_{13} = 20 \text{ V}$$

$$\omega_2 = 2 \cdot \omega = 2 \cdot 10^8 \frac{\text{rad}}{\text{s}}$$

$$I_{13} = \frac{20}{10} = 2 \text{ A}$$



$$Z_1 = Z_{L1} + Z_{C1}$$

$$Z_{L1} = j\omega L_1 = j \cdot 2 \cdot 10^8 \cdot 100 \cdot 10^{-9} = 20j \Omega$$

$$Z_{C1} = \frac{1}{j\omega C_1} = \frac{1}{j \cdot 2 \cdot 10^8 \cdot 10^{-9}} = -5j \Omega$$

$$Z_1 = 20j - 5j = 15j \Omega$$

$$Z_2 = \frac{R_1 R_3}{R_1 + R_3} = \frac{20}{3} \Omega$$

$$Z_3 = Z_1 \parallel Z_2 = \frac{Z_1 \cdot Z_2}{Z_1 + Z_2} \approx 5,57 + 2,47j \Omega$$

$$i_{r23} = I_1 \cdot \frac{Z_3}{Z_3 + R_2} \approx 0,45 + 0,15j \text{ [A]}$$

$$u_{r23} = i_{r23} \cdot R_2 = (0,45 + 0,15j) \cdot 20 = 9 + 3j \text{ [V]}$$

$$i_{r2} = 0,5 + 0 + 0,45 \cos(2 \cdot 10^8 t) - 0,15 \sin(2 \cdot 10^8 t) \text{ A}$$

$$u_{r2} = 10 + 0 + 9 \cos(2 \cdot 10^8 t) - 3 \sin(2 \cdot 10^8 t) \text{ V}$$

$$P = \sum \frac{1}{2} \frac{u^2}{R} = \frac{10^2}{20} + \frac{1}{2} \cdot \frac{9^2 + 3^2}{20} = 5 + 2,25 = 7,25 \text{ W}$$

$$P \text{ dla DC} = \frac{u^2}{R}, \quad \text{dla AC} \frac{1}{2} \frac{u^2}{R}$$

